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Personal Data:

Date of Birth:

13th July, 1993

Gender: Male

Nationality:
Indian

Marital Status:
Married

Languages Known:

Telugu, English, Hindi.

RANJITH KUMAR ENUGALA

Experience: 7+ Years as Power Electronics and Embedded Hardware design, PCB Design, Application Engineer

CAREER OBJECTIVE

Looking for a challenging role which demands the best of my professional ability in terms of technical and analytical skills and helps me in broadening and enhancing my current skill and knowledge.

PROFESSIONAL EXPERIENCE

1. Sagar Semiconductors Pvt Ltd.

- Working as Hardware design, Application Engineer for Power Electronic devices since July-2024.

Responsibilities:

- Design and validate power electronic circuits using Sagar MOSFETs for applications such as EV controllers, BMS, chargers, inverters, and BLDC motor controllers.
- Develop reference designs, evaluation boards, and test circuits showcasing MOSFET performance.
- Evaluate alternative MOSFET technologies (e.g., SiC, GaN) and support design migration.
- Work directly with customers to analyze and resolve design, testing, and field issues in which Sagar MOSFETs are used.
- Compare competitor MOSFETs vs Sagar MOSFETs and Provide application-based solutions gate driver tuning, snubber optimization, and protection circuit design.
- Coordinate with product and design teams to incorporate field feedback into new MOSFET designs.
- Support customer validation and reliability testing
- Validate MOSFETs under cranking, load dump, and short-circuit conditions.
- Generate comparison reports between Sagar and competitor parts.

2. Axonify Tech Systems Pvt Ltd.

- Worked as Embedded Hardware & AutoCAD Electrical Design Engineer for Electric Vehicle Chargers from Dec-2022 to June-2024

Responsibilities:

- Product development, design and releases for production of “AC EV chargers”, “DC EV Fast Chargers” for Automobiles.
- Preparation of charger Electrical schematics and EV charger infra plans drawings in AutoCAD Electrical as per requirements.
- Designing and Implementation of Proto type for Microcontroller based charger

circuits using Altium and OrCAD.

- Preparation of specifications, validation and verification reports for various chargers and performing design validation & verification tests.
- Preparation of Installation, Site readiness documents and Operation manuals for AC and DC EV chargers.
- Designing of Wiring harness drawings for EV chargers, Electric vehicle internal routings, site planning from panel boards to EV charger locations.
- PCB testing, Planning of Test jigs and fixtures, System level validation, resolving issues received from sites, preparation of Test reports, Resolving production issues and release ECNs.
- Reviewing standards and preparing project documents.

3. HBL Power Systems Limited.

- Worked as System level Design, Application and AutoCAD Electrical Design Engineer for Battery Chargers from Dec-2020 to Nov-2022.

Responsibilities:

- Overseeing product development, design and releases for production of “Battery Chargers”, “DC Distribution Boards”, “Battery Monitoring System” and “Cell Boosters”.
- Executing projects by understanding client requirements and supporting them with proper and appropriate design options as per their needs.
- Resolving issues related to customer comments on the submitted documents.
- Preparation of specifications for various components.
- Preparation of SLDs and Multi-line Drawings with AutoCAD Electrical.
- Selection and sizing of transformer, switchgear and cables.
- Interacted with the production team to implement and debug a new manufacturing process to reduce time taken for production.
- Developing new design on Battery charger, involving certain sub-assemblies.
- Working on development and performance assessment of a new automation tool for pre-order to submit tender documents.

4. Variturn Electro Products Private Limited.

- Worked as Project and Testing Engineer for converters like (AC-DC, DC-DC) at Variturn Electro Products from Oct-2017 to Dec-2020.
- Have experience in EMI/EMC and Environmental tests for MIL grade and commercial grade power supplies.

Responsibilities:

- Having experience in testing and project engineering for converters like AC-DC, DC-DC in power electronics.
- R&D in projects according to customer requirements and to improve efficiency.
- Testing of PCBs, Converters and Power Supplies.
- Designing of Required test jigs and fixtures for testing.

- Manage and monitor the maintenance of lab and equipment periodically and ensure the proper functioning.
- Responsible for generating reports pertaining to the purchase orders, cost limitations in purchasing required components related to the project need and company ISO standards related to my working department.
- Preparation of various Technical Documents like Specification Document, QAP, ATP, PSEQ/SOP, MDI.

ACADEMIC QUALIFICATIONS

- Completed **Bachelor of Technology** in Electrical and Electronics Engineering from Vaagdevi Engineering College with aggregate of **73.50%** in the year of 2016.
- Completed **Intermediate** from Vignan Junior College with aggregate of **95.2%** in the year of 2012.
- Completed **SSC** from ZPSS High School with aggregate of **86.16%** in the year of 2010.

PROJECTS

1. **Design of starter battery BMS** for Sodium, Lithium Ion batteries replacement to Lead Acid Batteries for our client Sodian Energy, Amara Raja

Heater, Temperature control circuits for batteries.

Reference Design of 1.2KW EV charger to showcasing Sagar MOSFET performance compared to competitor MOSFETs like Infineon, Toshiba, NXP etc.

2. **Design of 3.3KW and 7.4KW 1-PH AC EV Chargers, 30KW to 240 KW DC fast Chargers** for our clients HPCL, MG motors, Olectra, TATA motors, commercial and residential apartments, Public and private EV chargers.

Preparation of Technical specifications, BOM, Embedded hardware design using Main Microcontroller, Communication Microcontroller with Bluetooth, Wifi facility.

Designing of Electrical schematics, Mechanical Enclosure planning, testing procedures, Project planning, production related documents as per customer standards, offered third party and customer inspection procedures.

3. **Design of Battery Charger (DC System) Switchgear Panels** for our clients PDO, OETC, SEWA, KOC, Kuwait National Guard, PGCB, L&T, Shell Energy, ABB, NTPC, HPCL, IOCL, DMRC, CMRL, TATA Projects.

4. **Testing of Battery Monitoring System** for our clients Hitachi, NTPC, BHEL, ISGEC, Hindalco, Standardization, targeted to eliminate errors in post-order engineering, designed standard SLDs and Multi-line Drawings

5. **Quad Power Supply Unit as Project Engineer:** This is a 370W AC to DC Quad Output Power Supply with redundant Output.

6. 440V 3Phase AC-DC Converters, DC-DC Converters as Project Engineer: This is 3 varieties of DC-DC and 3 Phase 440V AC-DC Converters (with the use of TMS320F28379D controller) which was qualified for the EMI EMC, MIL STD 461E, Environmental JSS 55555, 30G Shock Test.

TECHNICAL SKILLS

- Altium, Kicad, AutoCAD Electrical, OrCAD, E-Plan
- MS Office, PGDCA, SAP ERP-Material Management, SAP PP.
- RFSIM99, LTSPICE

SKILLS AND ABILITIES

- Team Management
- Problem Solving Skills
- Self-motivated & Quick Learning
- Product testing
- Bugs Identification

DECLARATION

I hereby ensure that the above-mentioned information is true and correct to the best of my knowledge and belief.

Place: Hyderabad

Date:

RANJITH KUMAR E.